Stat441_FprojPrepExploration_HartC

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1 Statistical Modeling of the Old School Runescape Highscores

2 Preparation and Exploration

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2.3 Statistical Machine Learning I - Fall 2024

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3 Imports

3.0.1 Goal

Research Question: Can you predict a player's total xp based on the "kill" count of each in-game boss, despite the extremely sparse data?

[1]:
```
import pandas as pd
import random
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

4 Data Source

https://github.com/lukearend/osrs-hiscores

Using the player-stats-raw.csv 1.5gbs

"Data were scraped for the top 2 million players on the OSRS hiscores. Data consists of xp, rank, and level in each OSRS skill and overall, along with rank and score stats for clue scrolls, minigames and bosses. Account data was then deduplicated, sorted." Luke Arend on Github

[126]:
```
df = pd.read_csv('player-stats-raw.csv')
df # 2007339 rows × 198 columns
```

[126]:

	username	total_rank	total_level	total_xp	attack_rank	\
0	Lynx Titan	1	2277	4600000000	15	
1	Hey Jase	2	2277	4600000000	21	

```
2            ShawnBay           3         2277  4600000000          120
3               senZe           4         2277  4600000000          113
4            Tomdabom           5         2277  4600000000          151
…                 …          …           …          …          …
2007334  420shortribs     1999996          807     1560053           -1
2007335       Luke262     1999997          807     1520618      1739960
2007336    Rer Eirikr     1999998          807     1510324           -1
2007337  Onyxius Void     1999999          807     1389232           -1
2007338         Y J C     2000000          807     1366785           -1

         attack_level    attack_xp  defence_rank  defence_level   defence_xp  \
0                  99  200000000.0            28             99  200000000.0
1                  99  200000000.0            60             99  200000000.0
2                  99  200000000.0            66             99  200000000.0
3                  99  200000000.0            85             99  200000000.0
4                  99  200000000.0            74             99  200000000.0
…                 …           …           …              …          …
2007334            54     162301.0       1858558             51     119253.0
2007335            60     291549.0       1999554             49      95514.0
2007336            40      38843.0            -1             35      23994.0
2007337            50     111864.0       1943065             50     102306.0
2007338            52     125600.0            -1             45      64792.0

         strength_rank  strength_level  strength_xp  hitpoints_rank  \
0                   18              99  200000000.0               7
1                   65              99  200000000.0              11
2                   50              99  200000000.0              56
3                   94              99  200000000.0              73
4                  102              99  200000000.0              62
…                 …              …           …              …
2007334             -1              52     132354.0              -1
2007335             -1              52     124072.0              -1
2007336             -1              38      33447.0              -1
2007337             -1              51     118024.0              -1
2007338             -1              50     111553.0              -1

         hitpoints_level  hitpoints_xp  ranged_rank  ranged_level  \
0                     99     200000000            8            99
1                     99     200000000           11            99
2                     99     200000000           43            99
3                     99     200000000           80            99
4                     99     200000000           60            99
…                    …          …           …           …
2007334               56        184501           -1            50
2007335               53        140842           -1            26
2007336               27         10540           -1            13
2007337               48         87886           -1            30
```

```
2007338              53        142383           -1            50

           ranged_xp  prayer_rank  prayer_level    prayer_xp  magic_rank  \
0        200000000.0           11            99  200000000.0          32
1        200000000.0           16            99  200000000.0          42
2        200000000.0           25            99  200000000.0          34
3        200000000.0           57            99  200000000.0          48
4        200000000.0           66            99  200000000.0          37
…              …          …           …          …          …
2007334     104391.0           -1            42     46976.0          -1
2007335       9613.0      1869247            44     59206.0          -1
2007336       2012.0      1471978            50    106774.0          -1
2007337      14410.0           -1            36     26929.0          -1
2007338     109238.0      1498966            50    102137.0          -1

         magic_level     magic_xp  cooking_rank  cooking_level   cooking_xp  \
0                 99  200000000.0           159             99  200000000.0
1                 99  200000000.0            34             99  200000000.0
2                 99  200000000.0           222             99  200000000.0
3                 99  200000000.0            39             99  200000000.0
4                 99  200000000.0            62             99  200000000.0
…               …           …           …            …          …
2007334           43      52922.0            -1             46      71773.0
2007335           40      39759.0            -1             47      82453.0
2007336           43      50953.0       1982831             50     109758.0
2007337           50     110602.0            -1             41      43458.0
2007338           52     125967.0            -1             45      63115.0

         woodcutting_rank  woodcutting_level  woodcutting_xp  fletching_rank  \
0                      15                 99     200000000.0              12
1                      40                 99     200000000.0               8
2                     113                 99     200000000.0             105
3                     129                 99     200000000.0              15
4                     146                 99     200000000.0             100
…                   …                 …            …               …
2007334                -1                 51        119512.0         1704237
2007335                -1                 31         15730.0              -1
2007336                -1                 47         82817.0              -1
2007337                -1                 37         28972.0              -1
2007338                -1                 51        114315.0              -1

         fletching_level  fletching_xp  fishing_rank  fishing_level  \
0                     99   200000000.0             9             99
1                     99   200000000.0            66             99
2                     99   200000000.0            18             99
3                     99   200000000.0            49             99
4                     99   200000000.0            17             99
```

```
…                 …           …           …            …
2007334            44       57855.0          -1           44
2007335            14        2388.0          -1           48
2007336             8         869.0          -1           -1
2007337            30       14017.0          -1           43
2007338            12        1748.0          -1           45

          fishing_xp  firemaking_rank  firemaking_level  firemaking_xp  \
0        200000000.0              48                99    200000000.0
1        200000000.0              94                99    200000000.0
2        200000000.0             246                99    200000000.0
3        200000000.0             272                99    200000000.0
4        200000000.0             286                99    200000000.0
…              …               …                 …            …
2007334      59612.0              -1                 6          640.0
2007335      86637.0         1211368                62       341339.0
2007336         -1.0              -1                49        93350.0
2007337      53444.0              -1                30        14800.0
2007338      62880.0         1669333                50       109756.0

          crafting_rank  crafting_level  crafting_xp  smithing_rank  \
0                     4              99  200000000.0              3
1                     3              99  200000000.0              4
2                    28              99  200000000.0             91
3                    11              99  200000000.0             25
4                    45              99  200000000.0             70
…                 …               …           …            …
2007334              -1              36      26610.0        1440833
2007335              -1              24       7116.0             -1
2007336              -1              36      26868.0        1378092
2007337         1881533              47      76324.0        1252542
2007338         1714029              50     108187.0        1844394

          smithing_level  smithing_xp  mining_rank  mining_level    mining_xp  \
0                     99  200000000.0           25            99  200000000.0
1                     99  200000000.0           37            99  200000000.0
2                     99  200000000.0           17            99  200000000.0
3                     99  200000000.0           27            99  200000000.0
4                     99  200000000.0           21            99  200000000.0
…                  …           …            …             …           …
2007334               50     105954.0      1344825            59     271766.0
2007335               39      34402.0           -1            45      66850.0
2007336               51     112263.0      1116458            62     338166.0
2007337               52     135778.0      1049355            63     394293.0
2007338               41      45187.0           -1            31      15842.0

          herblore_rank  …  kraken_score  kreearra_rank  kreearra_score  \
```

```
0                    5  …            -1             -1             -1
1                    6  …          2612            556           4110
2                   21  …            -1             -1             -1
3                    9  …            -1             -1             -1
4                   34  …           652             -1             -1
…                  …  …           …              …              …
2007334             -1  …            -1             -1             -1
2007335        1725099  …            -1             -1             -1
2007336        1837844  …            -1             -1             -1
2007337        1827199  …            -1             -1             -1
2007338        1514589  …            -1             -1             -1

         kril_tsutsaroth_rank  kril_tsutsaroth_score  mimic_rank  mimic_score  \
0                          -1                     -1          -1           -1
1                        2162                   1250           3          274
2                          -1                     -1          -1           -1
3                       90060                     72          -1           -1
4                      104237                     56        7506            9
…                         …                     …          …          …
2007334                    -1                     -1          -1           -1
2007335                    -1                     -1          -1           -1
2007336                    -1                     -1          -1           -1
2007337                    -1                     -1          -1           -1
2007338                    -1                     -1          -1           -1

         nex_rank  nex_score  nightmare_rank  nightmare_score  \
0              -1         -1              -1               -1
1            2404       2376             847             2243
2              -1         -1              -1               -1
3              -1         -1              -1               -1
4           23779        403              -1               -1
…             …        …              …              …
2007334        -1         -1              -1               -1
2007335        -1         -1              -1               -1
2007336        -1         -1              -1               -1
2007337        -1         -1              -1               -1
2007338        -1         -1              -1               -1

         phosanis_nightmare_rank  phosanis_nightmare_score  obor_rank  \
0                             -1                        -1         -1
1                            916                       831     132978
2                             -1                        -1         -1
3                             -1                        -1         -1
4                             -1                        -1         -1
…                            …                       …        …
2007334                       -1                        -1         -1
2007335                       -1                        -1         -1
```

```
2007336                      -1                        -1         -1
2007337                      -1                        -1         -1
2007338                      -1                        -1         -1

         obor_score  sarachnis_rank  sarachnis_score  scorpia_rank  \
0                -1              -1               -1            -1
1                 6             705             3144          3307
2                -1              -1               -1            -1
3                -1              -1               -1            -1
4                -1              -1               -1            -1
…               …               …                …             …
2007334          -1              -1               -1            -1
2007335          -1              -1               -1            -1
2007336          -1              -1               -1            -1
2007337          -1              -1               -1            -1
2007338          -1              -1               -1            -1

         scorpia_score  skotizo_rank  skotizo_score  tempoross_rank  \
0                   -1            -1             -1              -1
1                  766         55766             41           46408
2                   -1            -1             -1              -1
3                   -1            -1             -1              -1
4                   -1            45            617              -1
…                  …             …              …               …
2007334             -1            -1             -1              -1
2007335             -1            -1             -1              -1
2007336             -1            -1             -1              -1
2007337             -1            -1             -1              -1
2007338             -1            -1             -1              -1

         tempoross_score  the_gauntlet_rank  the_gauntlet_score  \
0                     -1                 -1                  -1
1                    150                 -1                  -1
2                     -1                 -1                  -1
3                     -1                 -1                  -1
4                     -1                 -1                  -1
…                    …                  …                   …
2007334               -1                 -1                  -1
2007335               -1                 -1                  -1
2007336               -1                 -1                  -1
2007337               -1                 -1                  -1
2007338               -1                 -1                  -1

         the_corrupted_gauntlet_rank  the_corrupted_gauntlet_score  \
0                                 -1                            -1
1                              78646                            10
2                                 -1                            -1
```

```
3                                  -1                                  -1
4                               38232                                 137
…                                   …                                   …
2007334                            -1                                  -1
2007335                            -1                                  -1
2007336                            -1                                  -1
2007337                            -1                                  -1
2007338                            -1                                  -1

         theatre_of_blood_rank  theatre_of_blood_score  \
0                           -1                      -1
1                         7361                     661
2                           -1                      -1
3                           -1                      -1
4                           -1                      -1
…                           …                       …
2007334                     -1                      -1
2007335                     -1                      -1
2007336                     -1                      -1
2007337                     -1                      -1
2007338                     -1                      -1

         theatre_of_blood_hard_mode_rank  theatre_of_blood_hard_mode_score  \
0                                     -1                                -1
1                                   2341                               189
2                                     -1                                -1
3                                     -1                                -1
4                                     -1                                -1
…                                     …                                 …
2007334                               -1                                -1
2007335                               -1                                -1
2007336                               -1                                -1
2007337                               -1                                -1
2007338                               -1                                -1

         thermonuclear_smoke_devil_rank  thermonuclear_smoke_devil_score  \
0                                    -1                               -1
1                                  2843                             3797
2                                    -1                               -1
3                                    -1                               -1
4                                 36955                             1020
…                                     …                                …
2007334                              -1                               -1
2007335                              -1                               -1
2007336                              -1                               -1
2007337                              -1                               -1
2007338                              -1                               -1
```

	tzkal_zuk_rank	tzkal_zuk_score	tztok_jad_rank	tztok_jad_score	\
0	-1	-1	143	186	
1	2447	10	168	175	
2	-1	-1	-1	-1	
3	-1	-1	18787	23	
4	13052	1	19186	23	
…	…	…	…	…	
2007334	-1	-1	-1	-1	
2007335	-1	-1	-1	-1	
2007336	-1	-1	-1	-1	
2007337	-1	-1	-1	-1	
2007338	-1	-1	-1	-1	

	venenatis_rank	venenatis_score	vetion_rank	vetion_score	\
0	-1	-1	-1	-1	
1	12299	447	372	2274	
2	-1	-1	-1	-1	
3	-1	-1	-1	-1	
4	-1	-1	20215	87	
…	…	…	…	…	
2007334	-1	-1	-1	-1	
2007335	-1	-1	-1	-1	
2007336	-1	-1	-1	-1	
2007337	-1	-1	-1	-1	
2007338	-1	-1	-1	-1	

	vorkath_rank	vorkath_score	wintertodt_rank	wintertodt_score	\
0	-1	-1	-1	-1	
1	3342	5172	126171	542	
2	4454	4333	584	3858	
3	-1	-1	1395	2064	
4	43059	1076	567780	94	
…	…	…	…	…	
2007334	-1	-1	-1	-1	
2007335	-1	-1	-1	-1	
2007336	-1	-1	-1	-1	
2007337	-1	-1	-1	-1	
2007338	-1	-1	-1	-1	

	zalcano_rank	zalcano_score	zulrah_rank	zulrah_score	\
0	-1	-1	-1	-1	
1	45528	150	1635	10518	
2	-1	-1	-1	-1	
3	-1	-1	-1	-1	
4	-1	-1	35149	2010	
…	…	…	…	…	

```
2007334           -1           -1           -1           -1
2007335           -1           -1           -1           -1
2007336           -1           -1           -1           -1
2007337           -1           -1           -1           -1
2007338           -1           -1           -1           -1

                                ts
0        2022-07-21T05:29:27.357868
1        2022-07-21T05:29:27.310638
2        2022-07-21T05:29:27.324108
3        2022-07-21T05:29:27.454984
4        2022-07-21T05:29:27.411068
…                               …
2007334  2022-07-22T01:05:32.077508
2007335  2022-07-22T01:05:32.077159
2007336  2022-07-22T01:05:32.052940
2007337  2022-07-22T01:05:32.119929
2007338  2022-07-22T01:05:32.048250

[2007339 rows x 198 columns]
```

[4]:
```python
df = pd.read_csv('player-stats.csv')
df # 1999625 rows × 26 columns
```

4.1 Goals

1) Predict total xp based on boss kc
2) Predict barrows kill count based on xp in each skill
3) bin total_rank into 5 categories and then predict categoreis.

5 Data Exploration

[7]:
```python
# Load the dataset again if needed.
df = pd.read_csv('player-stats-raw.csv')
```

[8]:
```python
# Ensuring reproducibility by setting a random seed
random.seed(42)

# Generate three random selections of 10,000 rows
for i in range(1, 4):
    sample_df = df.sample(n=10000, random_state=random.randint(0, 10000))  #␣
 ↪Random seed changes for each file
    sample_df.to_csv(f'player_stats_sample_{i}.csv', index=False)
    print(f"Sample {i} saved as 'player_stats_sample_{i}.csv'")
```

```
Sample 1 saved as 'player_stats_sample_1.csv'
Sample 2 saved as 'player_stats_sample_2.csv'
```

```
Sample 3 saved as 'player_stats_sample_3.csv'
```

[9]:
```
# Load the sampled datasets
df1 = pd.read_csv('player_stats_sample_1.csv')
df2 = pd.read_csv('player_stats_sample_2.csv')
df3 = pd.read_csv('player_stats_sample_3.csv')
```

[10]:
```
df1.describe(include = "all")
```

[10]:

	username	total_rank	total_level	total_xp	attack_rank	\
count	10000	1.000000e+04	10000.00000	1.000000e+04	1.000000e+04	
unique	9999	NaN	NaN	NaN	NaN	
top	Hiddy	NaN	NaN	NaN	NaN	
freq	2	NaN	NaN	NaN	NaN	
mean	NaN	1.005892e+06	1424.67010	6.233747e+07	7.853244e+05	
std	NaN	5.811232e+05	403.35253	1.237551e+08	6.040859e+05	
min	NaN	2.400000e+01	807.00000	8.689970e+05	-1.000000e+00	
25%	NaN	5.030038e+05	1065.75000	8.746008e+06	2.143605e+05	
50%	NaN	1.012052e+06	1392.50000	2.465564e+07	7.430035e+05	
75%	NaN	1.509058e+06	1753.00000	7.036275e+07	1.281353e+06	
max	NaN	1.999952e+06	2277.00000	4.600000e+09	1.999897e+06	

	attack_level	attack_xp	defence_rank	defence_level	defence_xp	\
count	10000.000000	9.997000e+03	1.000000e+04	10000.000000	9.982000e+03	
unique	NaN	NaN	NaN	NaN	NaN	
top	NaN	NaN	NaN	NaN	NaN	
freq	NaN	NaN	NaN	NaN	NaN	
mean	68.010900	4.071134e+06	7.297442e+05	63.969200	3.468804e+06	
std	29.440483	8.122788e+06	5.905013e+05	31.536755	6.977748e+06	
min	-1.000000	-1.000000e+00	-1.000000e+00	-1.000000	-1.000000e+00	
25%	61.000000	3.235830e+05	1.523065e+05	60.000000	2.738282e+05	
50%	75.000000	1.213460e+06	6.859375e+05	71.000000	8.695325e+05	
75%	87.000000	4.013912e+06	1.202393e+06	85.000000	3.352316e+06	
max	99.000000	2.000000e+08	1.999682e+06	99.000000	2.000000e+08	

	…	vetion_score	vorkath_rank	vorkath_score	wintertodt_rank	\
count	…	10000.000000	10000.000000	10000.000000	10000.000000	
unique	…	NaN	NaN	NaN	NaN	
top	…	NaN	NaN	NaN	NaN	
freq	…	NaN	NaN	NaN	NaN	
mean	…	3.315900	31100.184600	101.498900	117677.257700	
std	…	96.499287	78877.892457	575.558318	208601.135626	
min	…	-1.000000	-1.000000	-1.000000	-1.000000	
25%	…	-1.000000	-1.000000	-1.000000	-1.000000	
50%	…	-1.000000	-1.000000	-1.000000	-1.000000	
75%	…	-1.000000	-1.000000	-1.000000	164400.250000	
max	…	6462.000000	344688.000000	30780.000000	739989.000000	

```
        wintertodt_score  zalcano_rank  zalcano_score    zulrah_rank  \
count       10000.000000  10000.000000   10000.000000   10000.000000
unique               NaN           NaN            NaN            NaN
top                  NaN           NaN            NaN            NaN
freq                 NaN           NaN            NaN            NaN
mean           92.108400   2363.066300      12.988400   17171.597400
std           226.713464  12274.176228     130.196114   51576.349066
min            -1.000000     -1.000000      -1.000000      -1.000000
25%            -1.000000     -1.000000      -1.000000      -1.000000
50%            -1.000000     -1.000000      -1.000000      -1.000000
75%            98.000000     -1.000000      -1.000000      -1.000000
max          8516.000000  96339.000000    6067.000000  257089.000000

        zulrah_score                          ts
count    10000.00000                       10000
unique           NaN                       10000
top              NaN  2022-07-21T21:39:06.320583
freq             NaN                           1
mean       155.18540                         NaN
std        930.93639                         NaN
min         -1.00000                         NaN
25%         -1.00000                         NaN
50%         -1.00000                         NaN
75%         -1.00000                         NaN
max      41751.00000                         NaN

[11 rows x 198 columns]
```

[11]:
```python
# Force display of all columns in pd.info
with pd.option_context('display.max_info_columns', 200):
    df1.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 198 columns):
 #    Column                                Non-Null Count  Dtype
---   ------                                --------------  -----
 0    username                              10000 non-null  object
 1    total_rank                            10000 non-null  int64
 2    total_level                           10000 non-null  int64
 3    total_xp                              10000 non-null  int64
 4    attack_rank                           10000 non-null  int64
 5    attack_level                          10000 non-null  int64
 6    attack_xp                             9997 non-null   float64
 7    defence_rank                          10000 non-null  int64
 8    defence_level                         10000 non-null  int64
 9    defence_xp                            9982 non-null   float64
```

```
10   strength_rank                10000 non-null  int64
11   strength_level               10000 non-null  int64
12   strength_xp                  9999 non-null   float64
13   hitpoints_rank               10000 non-null  int64
14   hitpoints_level              10000 non-null  int64
15   hitpoints_xp                 10000 non-null  int64
16   ranged_rank                  10000 non-null  int64
17   ranged_level                 10000 non-null  int64
18   ranged_xp                    10000 non-null  float64
19   prayer_rank                  10000 non-null  int64
20   prayer_level                 10000 non-null  int64
21   prayer_xp                    9996 non-null   float64
22   magic_rank                   10000 non-null  int64
23   magic_level                  10000 non-null  int64
24   magic_xp                     10000 non-null  float64
25   cooking_rank                 10000 non-null  int64
26   cooking_level                10000 non-null  int64
27   cooking_xp                   10000 non-null  float64
28   woodcutting_rank             10000 non-null  int64
29   woodcutting_level            10000 non-null  int64
30   woodcutting_xp               10000 non-null  float64
31   fletching_rank               10000 non-null  int64
32   fletching_level              10000 non-null  int64
33   fletching_xp                 9986 non-null   float64
34   fishing_rank                 10000 non-null  int64
35   fishing_level                10000 non-null  int64
36   fishing_xp                   10000 non-null  float64
37   firemaking_rank              10000 non-null  int64
38   firemaking_level             10000 non-null  int64
39   firemaking_xp                10000 non-null  float64
40   crafting_rank                10000 non-null  int64
41   crafting_level               10000 non-null  int64
42   crafting_xp                  10000 non-null  float64
43   smithing_rank                10000 non-null  int64
44   smithing_level               10000 non-null  int64
45   smithing_xp                  10000 non-null  float64
46   mining_rank                  10000 non-null  int64
47   mining_level                 10000 non-null  int64
48   mining_xp                    10000 non-null  float64
49   herblore_rank                10000 non-null  int64
50   herblore_level               10000 non-null  int64
51   herblore_xp                  9969 non-null   float64
52   agility_rank                 10000 non-null  int64
53   agility_level                10000 non-null  int64
54   agility_xp                   9989 non-null   float64
55   thieving_rank                10000 non-null  int64
56   thieving_level               10000 non-null  int64
57   thieving_xp                  9989 non-null   float64
```

```
58   slayer_rank                    10000 non-null  int64
59   slayer_level                   10000 non-null  int64
60   slayer_xp                      9981 non-null   float64
61   farming_rank                   10000 non-null  int64
62   farming_level                  10000 non-null  int64
63   farming_xp                     9951 non-null   float64
64   runecraft_rank                 10000 non-null  int64
65   runecraft_level                10000 non-null  int64
66   runecraft_xp                   9920 non-null   float64
67   hunter_rank                    10000 non-null  int64
68   hunter_level                   10000 non-null  int64
69   hunter_xp                      9923 non-null   float64
70   construction_rank              10000 non-null  int64
71   construction_level             10000 non-null  int64
72   construction_xp                9928 non-null   float64
73   unknown_rank                   10000 non-null  int64
74   unknown_score                  10000 non-null  int64
75   bounty_hunter_hunter_rank      10000 non-null  int64
76   bounty_hunter_hunter_score     10000 non-null  int64
77   bounty_hunter_rogue_rank       10000 non-null  int64
78   bounty_hunter_rogue_score      10000 non-null  int64
79   clue_scrolls_all_rank          10000 non-null  int64
80   clue_scrolls_all_score         10000 non-null  int64
81   clue_scrolls_beginner_rank     10000 non-null  int64
82   clue_scrolls_beginner_score    10000 non-null  int64
83   clue_scrolls_easy_rank         10000 non-null  int64
84   clue_scrolls_easy_score        10000 non-null  int64
85   clue_scrolls_medium_rank       10000 non-null  int64
86   clue_scrolls_medium_score      10000 non-null  int64
87   clue_scrolls_hard_rank         10000 non-null  int64
88   clue_scrolls_hard_score        10000 non-null  int64
89   clue_scrolls_elite_rank        10000 non-null  int64
90   clue_scrolls_elite_score       10000 non-null  int64
91   clue_scrolls_master_rank       10000 non-null  int64
92   clue_scrolls_master_score      10000 non-null  int64
93   lms_rank_rank                  10000 non-null  int64
94   lms_rank_score                 10000 non-null  int64
95   pvp_arena_rank                 10000 non-null  int64
96   pvp_arena_score                10000 non-null  int64
97   soul_wars_zeal_rank            10000 non-null  int64
98   soul_wars_zeal_score           10000 non-null  int64
99   rifts_closed_rank              10000 non-null  int64
100  rifts_closed_score             10000 non-null  int64
101  abyssal_sire_rank              10000 non-null  int64
102  abyssal_sire_score             10000 non-null  int64
103  alchemical_hydra_rank          10000 non-null  int64
104  alchemical_hydra_score         10000 non-null  int64
105  barrows_chests_rank            10000 non-null  int64
```

```
106  barrows_chests_score                      10000 non-null  int64
107  bryophyta_rank                            10000 non-null  int64
108  bryophyta_score                           10000 non-null  int64
109  callisto_rank                             10000 non-null  int64
110  callisto_score                            10000 non-null  int64
111  cerberus_rank                             10000 non-null  int64
112  cerberus_score                            10000 non-null  int64
113  chambers_of_xeric_rank                    10000 non-null  int64
114  chambers_of_xeric_score                   10000 non-null  int64
115  chambers_of_xeric_challenge_mode_rank     10000 non-null  int64
116  chambers_of_xeric_challenge_mode_score    10000 non-null  int64
117  chaos_elemental_rank                      10000 non-null  int64
118  chaos_elemental_score                     10000 non-null  int64
119  chaos_fanatic_rank                        10000 non-null  int64
120  chaos_fanatic_score                       10000 non-null  int64
121  commander_zilyana_rank                    10000 non-null  int64
122  commander_zilyana_score                   10000 non-null  int64
123  corporeal_beast_rank                      10000 non-null  int64
124  corporeal_beast_score                     10000 non-null  int64
125  crazy_archaeologist_rank                  10000 non-null  int64
126  crazy_archaeologist_score                 10000 non-null  int64
127  dagannoth_prime_rank                      10000 non-null  int64
128  dagannoth_prime_score                     10000 non-null  int64
129  dagannoth_rex_rank                        10000 non-null  int64
130  dagannoth_rex_score                       10000 non-null  int64
131  dagannoth_supreme_rank                    10000 non-null  int64
132  dagannoth_supreme_score                   10000 non-null  int64
133  deranged_archaeologist_rank               10000 non-null  int64
134  deranged_archaeologist_score              10000 non-null  int64
135  general_graardor_rank                     10000 non-null  int64
136  general_graardor_score                    10000 non-null  int64
137  giant_mole_rank                           10000 non-null  int64
138  giant_mole_score                          10000 non-null  int64
139  grotesque_guardians_rank                  10000 non-null  int64
140  grotesque_guardians_score                 10000 non-null  int64
141  hespori_rank                              10000 non-null  int64
142  hespori_score                             10000 non-null  int64
143  kalphite_queen_rank                       10000 non-null  int64
144  kalphite_queen_score                      10000 non-null  int64
145  king_black_dragon_rank                    10000 non-null  int64
146  king_black_dragon_score                   10000 non-null  int64
147  kraken_rank                               10000 non-null  int64
148  kraken_score                              10000 non-null  int64
149  kreearra_rank                             10000 non-null  int64
150  kreearra_score                            10000 non-null  int64
151  kril_tsutsaroth_rank                      10000 non-null  int64
152  kril_tsutsaroth_score                     10000 non-null  int64
153  mimic_rank                                10000 non-null  int64
```

```
 154  mimic_score                          10000 non-null  int64
 155  nex_rank                             10000 non-null  int64
 156  nex_score                            10000 non-null  int64
 157  nightmare_rank                       10000 non-null  int64
 158  nightmare_score                      10000 non-null  int64
 159  phosanis_nightmare_rank              10000 non-null  int64
 160  phosanis_nightmare_score             10000 non-null  int64
 161  obor_rank                            10000 non-null  int64
 162  obor_score                           10000 non-null  int64
 163  sarachnis_rank                       10000 non-null  int64
 164  sarachnis_score                      10000 non-null  int64
 165  scorpia_rank                         10000 non-null  int64
 166  scorpia_score                        10000 non-null  int64
 167  skotizo_rank                         10000 non-null  int64
 168  skotizo_score                        10000 non-null  int64
 169  tempoross_rank                       10000 non-null  int64
 170  tempoross_score                      10000 non-null  int64
 171  the_gauntlet_rank                    10000 non-null  int64
 172  the_gauntlet_score                   10000 non-null  int64
 173  the_corrupted_gauntlet_rank          10000 non-null  int64
 174  the_corrupted_gauntlet_score         10000 non-null  int64
 175  theatre_of_blood_rank                10000 non-null  int64
 176  theatre_of_blood_score               10000 non-null  int64
 177  theatre_of_blood_hard_mode_rank      10000 non-null  int64
 178  theatre_of_blood_hard_mode_score     10000 non-null  int64
 179  thermonuclear_smoke_devil_rank       10000 non-null  int64
 180  thermonuclear_smoke_devil_score      10000 non-null  int64
 181  tzkal_zuk_rank                       10000 non-null  int64
 182  tzkal_zuk_score                      10000 non-null  int64
 183  tztok_jad_rank                       10000 non-null  int64
 184  tztok_jad_score                      10000 non-null  int64
 185  venenatis_rank                       10000 non-null  int64
 186  venenatis_score                      10000 non-null  int64
 187  vetion_rank                          10000 non-null  int64
 188  vetion_score                         10000 non-null  int64
 189  vorkath_rank                         10000 non-null  int64
 190  vorkath_score                        10000 non-null  int64
 191  wintertodt_rank                      10000 non-null  int64
 192  wintertodt_score                     10000 non-null  int64
 193  zalcano_rank                         10000 non-null  int64
 194  zalcano_score                        10000 non-null  int64
 195  zulrah_rank                          10000 non-null  int64
 196  zulrah_score                         10000 non-null  int64
 197  ts                                   10000 non-null  object
dtypes: float64(22), int64(174), object(2)
memory usage: 15.1+ MB
```

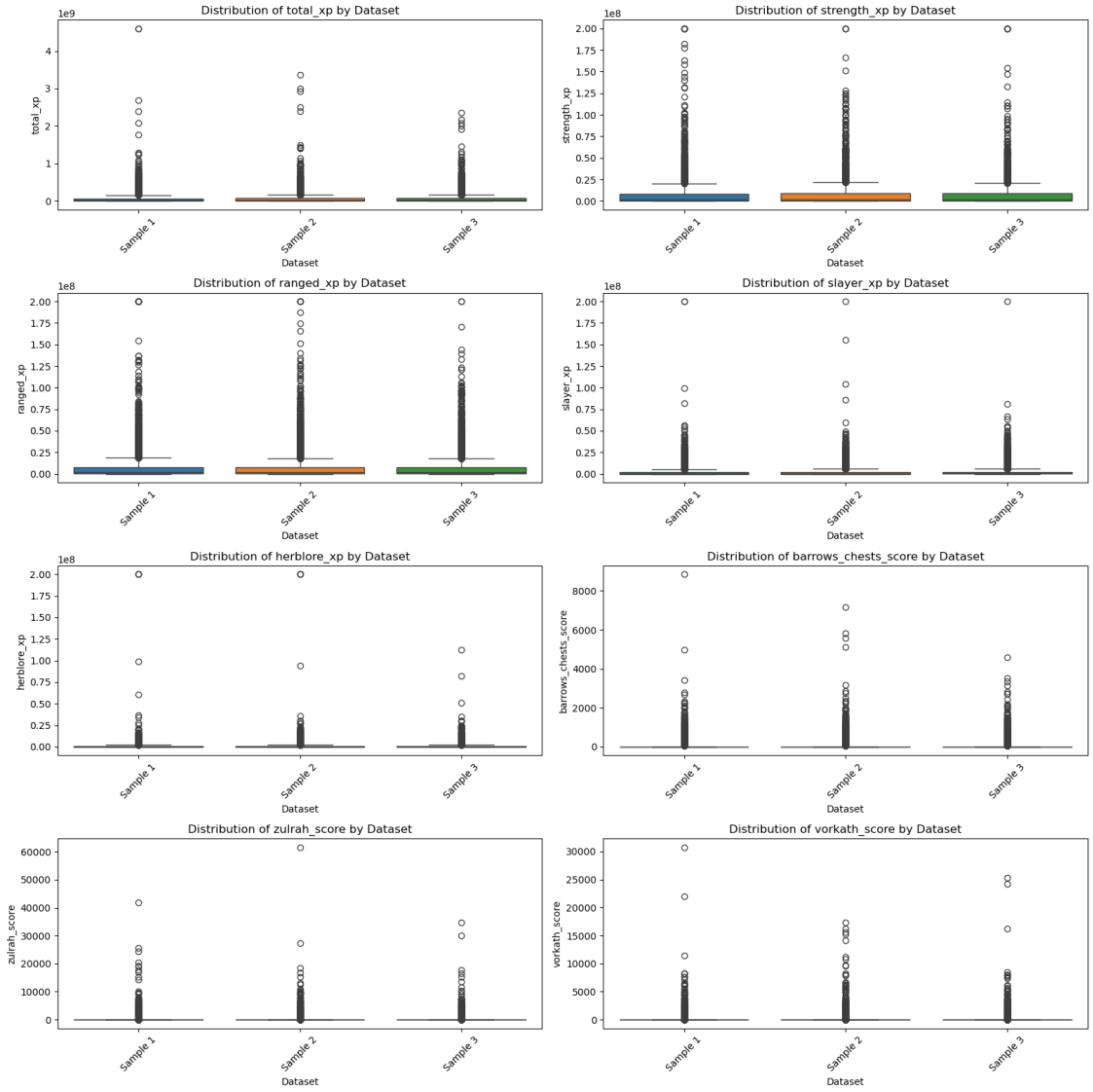
[12]:
```
# Combine the datasets into one with a new column to indicate the source
df1['Dataset'] = 'Sample 1'
df2['Dataset'] = 'Sample 2'
df3['Dataset'] = 'Sample 3'
combined_df = pd.concat([df1, df2, df3], axis=0)

# Reset the index to avoid duplicate index issues
combined_df.reset_index(drop=True, inplace=True)

# Columns to visualize
columns_to_plot = ['total_xp', 'strength_xp', 'ranged_xp', 'slayer_xp',␣
 ↪'herblore_xp',
                   'barrows_chests_score', 'zulrah_score', 'vorkath_score']

# Create a grid of plots
plt.figure(figsize=(16, 16))
for i, col in enumerate(columns_to_plot, 1):
    plt.subplot(4, 2, i)
    sns.boxplot(data=combined_df, x='Dataset', y=col, hue='Dataset',␣
 ↪dodge=False)
    plt.title(f'Distribution of {col} by Dataset')
    plt.xticks(rotation=45)
    plt.tight_layout()

# Show all plots
plt.show()
```



[13]:
```
plt.figure(figsize=(16, 16))
for i, col in enumerate(columns_to_plot, 1):
    plt.subplot(4, 2, i)
    sns.violinplot(data=combined_df, x='Dataset', y=col, palette='Set2')
    plt.title(f'Distribution of {col} by Dataset')
    plt.xticks(rotation=45)
    plt.tight_layout()

plt.show()
```

```
C:\Users\caleb\AppData\Local\Temp\ipykernel_13036\3957816599.py:4:
FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.violinplot(data=combined_df, x='Dataset', y=col, palette='Set2')
C:\Users\caleb\AppData\Local\Temp\ipykernel_13036\3957816599.py:4:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.violinplot(data=combined_df, x='Dataset', y=col, palette='Set2')
C:\Users\caleb\AppData\Local\Temp\ipykernel_13036\3957816599.py:4:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.violinplot(data=combined_df, x='Dataset', y=col, palette='Set2')
C:\Users\caleb\AppData\Local\Temp\ipykernel_13036\3957816599.py:4:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.violinplot(data=combined_df, x='Dataset', y=col, palette='Set2')
C:\Users\caleb\AppData\Local\Temp\ipykernel_13036\3957816599.py:4:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.violinplot(data=combined_df, x='Dataset', y=col, palette='Set2')
C:\Users\caleb\AppData\Local\Temp\ipykernel_13036\3957816599.py:4:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

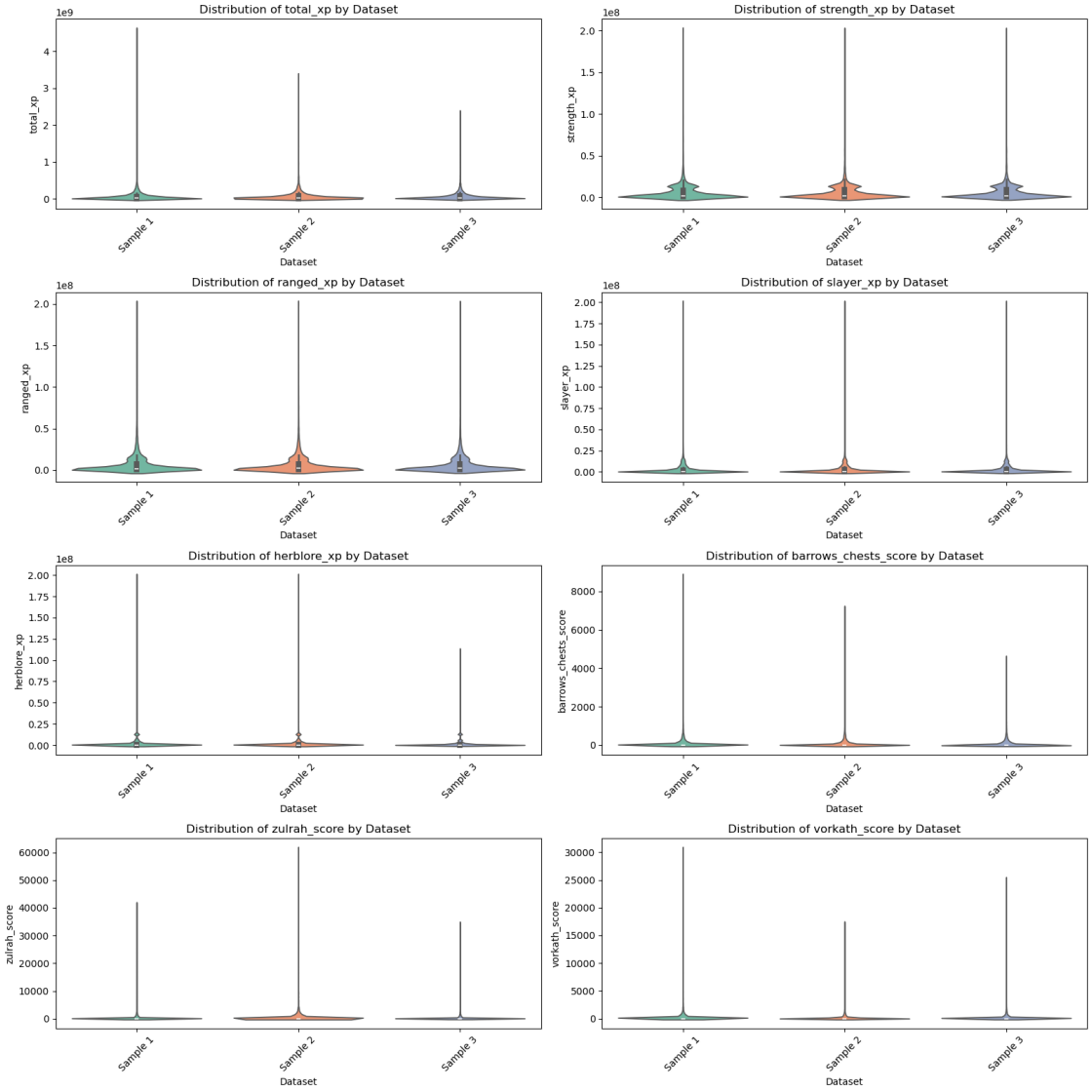
  sns.violinplot(data=combined_df, x='Dataset', y=col, palette='Set2')
C:\Users\caleb\AppData\Local\Temp\ipykernel_13036\3957816599.py:4:
FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.violinplot(data=combined_df, x='Dataset', y=col, palette='Set2')
C:\Users\caleb\AppData\Local\Temp\ipykernel_13036\3957816599.py:4:
FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in
v0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same
effect.

  sns.violinplot(data=combined_df, x='Dataset', y=col, palette='Set2')
```



6 Data Cleaning

Dropped all ‘rank’ columns

Dropped ‘timestamp’

Dropped ‘username’

Dropped ‘dataset’

Dropped all ‘level’ and ‘xp’ columns

Kept ‘total_xp’ and all columns with ‘score’ in them.

[38]:
```
dataframes = [df1, df2, df3]

# Loop through each dataframe in the list
for i, df in enumerate(dataframes, 1):
    # Drop all columns containing 'rank' in the name
    df = df.loc[:, ~df.columns.str.contains('rank')]

    # Drop the 'ts' column explicitly
    df = df.drop(columns=['ts'], errors='ignore')

    df = df.drop(columns=['Dataset'], errors='ignore')
    # Update the dataframe in the list
    dataframes[i-1] = df

    # Optionally: Verify the changes by checking the dataframe info
    print(f"Updated DataFrame {i} info:")
    df.info()
```

```
Updated DataFrame 1 info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Columns: 110 entries, username to zulrah_score
dtypes: float64(22), int64(87), object(1)
memory usage: 8.4+ MB
Updated DataFrame 2 info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Columns: 110 entries, username to zulrah_score
dtypes: float64(22), int64(87), object(1)
memory usage: 8.4+ MB
Updated DataFrame 3 info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Columns: 110 entries, username to zulrah_score
dtypes: float64(22), int64(87), object(1)
memory usage: 8.4+ MB
```

[40]:
```
df1 = dataframes[1]
df1.head()
```

[40]:

	username	total_level	total_xp	attack_level	attack_xp	\
0	Titus_M	1412	10883717	70	767255.0	
1	Fe Boof	1138	11967413	-1	-1.0	
2	Vieze Noob	1701	38709020	75	1219707.0	
3	0000 1 0000	1168	19987046	75	1326924.0	
4	BIGACLOWN	1428	19727936	83	2858189.0	

	defence_level	defence_xp	strength_level	strength_xp	hitpoints_level	\
0	68	646997.0	67	572443.0	69	
1	-1	-1.0	-1	-1.0	-1	
2	74	1133766.0	89	5031434.0	84	
3	70	778381.0	93	7768447.0	86	
4	70	768218.0	80	2186267.0	83	

	…	theatre_of_blood_hard_mode_score	thermonuclear_smoke_devil_score	\
0	…	-1	-1	
1	…	-1	-1	
2	…	-1	-1	
3	…	-1	-1	
4	…	-1	-1	

	tzkal_zuk_score	tztok_jad_score	venenatis_score	vetion_score	\
0	-1	-1	-1	-1	
1	-1	-1	-1	-1	
2	-1	-1	-1	-1	
3	-1	-1	-1	-1	
4	-1	-1	-1	-1	

	vorkath_score	wintertodt_score	zalcano_score	zulrah_score
0	-1	-1	-1	-1
1	-1	312	-1	-1
2	-1	69	-1	-1
3	-1	-1	-1	-1
4	-1	-1	-1	-1

[5 rows x 110 columns]

[42]:
```
with pd.option_context('display.max_info_columns', 200):
    df1.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10000 entries, 0 to 9999
Data columns (total 110 columns):
 #    Column                                  Non-Null Count  Dtype
---   ------                                  --------------  -----
```

```
0    username                10000 non-null  object
1    total_level             10000 non-null  int64
2    total_xp                10000 non-null  int64
3    attack_level            10000 non-null  int64
4    attack_xp               9997 non-null   float64
5    defence_level           10000 non-null  int64
6    defence_xp              9978 non-null   float64
7    strength_level          10000 non-null  int64
8    strength_xp             9997 non-null   float64
9    hitpoints_level         10000 non-null  int64
10   hitpoints_xp            10000 non-null  int64
11   ranged_level            10000 non-null  int64
12   ranged_xp               10000 non-null  float64
13   prayer_level            10000 non-null  int64
14   prayer_xp               9997 non-null   float64
15   magic_level             10000 non-null  int64
16   magic_xp                9999 non-null   float64
17   cooking_level           10000 non-null  int64
18   cooking_xp              10000 non-null  float64
19   woodcutting_level       10000 non-null  int64
20   woodcutting_xp          10000 non-null  float64
21   fletching_level         10000 non-null  int64
22   fletching_xp            9984 non-null   float64
23   fishing_level           10000 non-null  int64
24   fishing_xp              10000 non-null  float64
25   firemaking_level        10000 non-null  int64
26   firemaking_xp           10000 non-null  float64
27   crafting_level          10000 non-null  int64
28   crafting_xp             9999 non-null   float64
29   smithing_level          10000 non-null  int64
30   smithing_xp             10000 non-null  float64
31   mining_level            10000 non-null  int64
32   mining_xp               10000 non-null  float64
33   herblore_level          10000 non-null  int64
34   herblore_xp             9974 non-null   float64
35   agility_level           10000 non-null  int64
36   agility_xp              9988 non-null   float64
37   thieving_level          10000 non-null  int64
38   thieving_xp             9986 non-null   float64
39   slayer_level            10000 non-null  int64
40   slayer_xp               9984 non-null   float64
41   farming_level           10000 non-null  int64
42   farming_xp              9946 non-null   float64
43   runecraft_level         10000 non-null  int64
44   runecraft_xp            9904 non-null   float64
45   hunter_level            10000 non-null  int64
46   hunter_xp               9907 non-null   float64
47   construction_level      10000 non-null  int64
```

```
48   construction_xp                          9918 non-null   float64
49   unknown_score                            10000 non-null  int64
50   bounty_hunter_hunter_score               10000 non-null  int64
51   bounty_hunter_rogue_score                10000 non-null  int64
52   clue_scrolls_all_score                   10000 non-null  int64
53   clue_scrolls_beginner_score              10000 non-null  int64
54   clue_scrolls_easy_score                  10000 non-null  int64
55   clue_scrolls_medium_score                10000 non-null  int64
56   clue_scrolls_hard_score                  10000 non-null  int64
57   clue_scrolls_elite_score                 10000 non-null  int64
58   clue_scrolls_master_score                10000 non-null  int64
59   pvp_arena_score                          10000 non-null  int64
60   soul_wars_zeal_score                     10000 non-null  int64
61   rifts_closed_score                       10000 non-null  int64
62   abyssal_sire_score                       10000 non-null  int64
63   alchemical_hydra_score                   10000 non-null  int64
64   barrows_chests_score                     10000 non-null  int64
65   bryophyta_score                          10000 non-null  int64
66   callisto_score                           10000 non-null  int64
67   cerberus_score                           10000 non-null  int64
68   chambers_of_xeric_score                  10000 non-null  int64
69   chambers_of_xeric_challenge_mode_score   10000 non-null  int64
70   chaos_elemental_score                    10000 non-null  int64
71   chaos_fanatic_score                      10000 non-null  int64
72   commander_zilyana_score                  10000 non-null  int64
73   corporeal_beast_score                    10000 non-null  int64
74   crazy_archaeologist_score                10000 non-null  int64
75   dagannoth_prime_score                    10000 non-null  int64
76   dagannoth_rex_score                      10000 non-null  int64
77   dagannoth_supreme_score                  10000 non-null  int64
78   deranged_archaeologist_score             10000 non-null  int64
79   general_graardor_score                   10000 non-null  int64
80   giant_mole_score                         10000 non-null  int64
81   grotesque_guardians_score                10000 non-null  int64
82   hespori_score                            10000 non-null  int64
83   kalphite_queen_score                     10000 non-null  int64
84   king_black_dragon_score                  10000 non-null  int64
85   kraken_score                             10000 non-null  int64
86   kreearra_score                           10000 non-null  int64
87   kril_tsutsaroth_score                    10000 non-null  int64
88   mimic_score                              10000 non-null  int64
89   nex_score                                10000 non-null  int64
90   nightmare_score                          10000 non-null  int64
91   phosanis_nightmare_score                 10000 non-null  int64
92   obor_score                               10000 non-null  int64
93   sarachnis_score                          10000 non-null  int64
94   scorpia_score                            10000 non-null  int64
95   skotizo_score                            10000 non-null  int64
```

```
 96   tempoross_score                        10000 non-null  int64
 97   the_gauntlet_score                     10000 non-null  int64
 98   the_corrupted_gauntlet_score           10000 non-null  int64
 99   theatre_of_blood_score                 10000 non-null  int64
 100  theatre_of_blood_hard_mode_score       10000 non-null  int64
 101  thermonuclear_smoke_devil_score        10000 non-null  int64
 102  tzkal_zuk_score                        10000 non-null  int64
 103  tztok_jad_score                        10000 non-null  int64
 104  venenatis_score                        10000 non-null  int64
 105  vetion_score                           10000 non-null  int64
 106  vorkath_score                          10000 non-null  int64
 107  wintertodt_score                       10000 non-null  int64
 108  zalcano_score                          10000 non-null  int64
 109  zulrah_score                           10000 non-null  int64
dtypes: float64(22), int64(87), object(1)
memory usage: 8.4+ MB
```

[44]:
```
df1 = df1.loc[:, ~df.columns.str.contains('level')]
```

[66]:
```
df1 = df1.loc[:, ~((df1.columns.str.contains('xp')) & (df1.columns !=␣
↪'total_xp'))]
```

[70]:
```
pd.options.display.max_columns = 100
pd.options.display.max_rows = 100
df1.head()
```

[70]:

	username	total_xp	unknown_score	bounty_hunter_hunter_score	bounty_hunter_rogue_score	clue_scrolls_all_score	clue_scrolls_beginner_score	clue_scrolls_easy_score	clue_scrolls_medium_score	clue_scrolls_hard_score
0	Titus_M	10883717	-1	-1	-1	4	1	2		
1	Fe Boof	11967413	-1	-1	-1	-1	-1	-1		
2	Vieze Noob	38709020	-1	3	2	-1	-1	-1		
3	0000 1 0000	19987046	-1	7	9	14	1	7		
4	BIGACLOWN	19727936	-1	-1	-1	3	-1	-1		

```
0                         1                       -1
1                         1                       -1
2                        -1                       -1
3                         5                        1
4                         2                        1

   clue_scrolls_elite_score  clue_scrolls_master_score  pvp_arena_score  \
0                        -1                         -1               -1
1                        -1                         -1               -1
2                        -1                         -1               -1
3                        -1                         -1               -1
4                        -1                         -1               -1

   soul_wars_zeal_score  rifts_closed_score  abyssal_sire_score  \
0                    -1                  -1                  -1
1                    -1                  -1                  -1
2                    -1                  -1                  -1
3                    -1                  -1                  -1
4                    -1                  -1                  -1

   alchemical_hydra_score  barrows_chests_score  bryophyta_score  \
0                      -1                    -1               -1
1                      -1                    -1               -1
2                      -1                    -1               -1
3                      -1                    -1               -1
4                      -1                    -1               -1

   callisto_score  cerberus_score  chambers_of_xeric_score  \
0              -1              -1                       -1
1              -1              -1                       -1
2              -1              -1                       -1
3              -1              -1                       -1
4              -1              -1                       -1

   chambers_of_xeric_challenge_mode_score  chaos_elemental_score  \
0                                      -1                     -1
1                                      -1                     -1
2                                      -1                     -1
3                                      -1                     -1
4                                      -1                     -1

   chaos_fanatic_score  commander_zilyana_score  corporeal_beast_score  \
0                   -1                       -1                     -1
1                   -1                       -1                     -1
2                   -1                       -1                     -1
3                   -1                       -1                     -1
4                   -1                       -1                     -1
```

```
   crazy_archaeologist_score  dagannoth_prime_score  dagannoth_rex_score  \
0                         -1                     -1                   -1
1                         -1                     -1                   -1
2                         -1                     -1                   -1
3                         -1                     -1                   -1
4                         -1                     -1                   -1

   dagannoth_supreme_score  deranged_archaeologist_score  \
0                       -1                            -1
1                       -1                            -1
2                       -1                            -1
3                       -1                            -1
4                       -1                            -1

   general_graardor_score  giant_mole_score  grotesque_guardians_score  \
0                      -1                -1                         -1
1                      -1                -1                         -1
2                      -1                -1                         -1
3                      -1                -1                         -1
4                      -1                -1                         -1

   hespori_score  kalphite_queen_score  king_black_dragon_score  kraken_score  \
0             -1                    -1                       -1            -1
1             -1                    -1                       -1            -1
2             25                    -1                       50            -1
3             -1                    -1                       -1            -1
4             -1                    -1                       -1            -1

   kreearra_score  kril_tsutsaroth_score  mimic_score  nex_score  \
0              -1                     -1           -1         -1
1              -1                     -1           -1         -1
2              -1                     -1           -1         -1
3              -1                     -1           -1         -1
4              -1                     -1           -1         -1

   nightmare_score  phosanis_nightmare_score  obor_score  sarachnis_score  \
0               -1                        -1          -1               -1
1               -1                        -1          -1               -1
2               -1                        -1          -1               -1
3               -1                        -1           5               -1
4               -1                        -1          -1               -1

   scorpia_score  skotizo_score  tempoross_score  the_gauntlet_score  \
0             -1             -1               -1                  -1
1             -1             -1               -1                  -1
2             -1             -1               -1                  -1
```

```
3             -1             -1              -1                  -1
4             -1             -1              -1                  -1

   the_corrupted_gauntlet_score  theatre_of_blood_score  \
0                            -1                      -1
1                            -1                      -1
2                            -1                      -1
3                            -1                      -1
4                            -1                      -1

   theatre_of_blood_hard_mode_score  thermonuclear_smoke_devil_score  \
0                                -1                               -1
1                                -1                               -1
2                                -1                               -1
3                                -1                               -1
4                                -1                               -1

   tzkal_zuk_score  tztok_jad_score  venenatis_score  vetion_score  \
0               -1               -1               -1            -1
1               -1               -1               -1            -1
2               -1               -1               -1            -1
3               -1               -1               -1            -1
4               -1               -1               -1            -1

   vorkath_score  wintertodt_score  zalcano_score  zulrah_score
0             -1                -1             -1            -1
1             -1               312             -1            -1
2             -1                69             -1            -1
3             -1                -1             -1            -1
4             -1                -1             -1            -1
```

[74]:
```
df1 = df1.replace(-1, np.nan)
```

7 Sparsity Heatmap

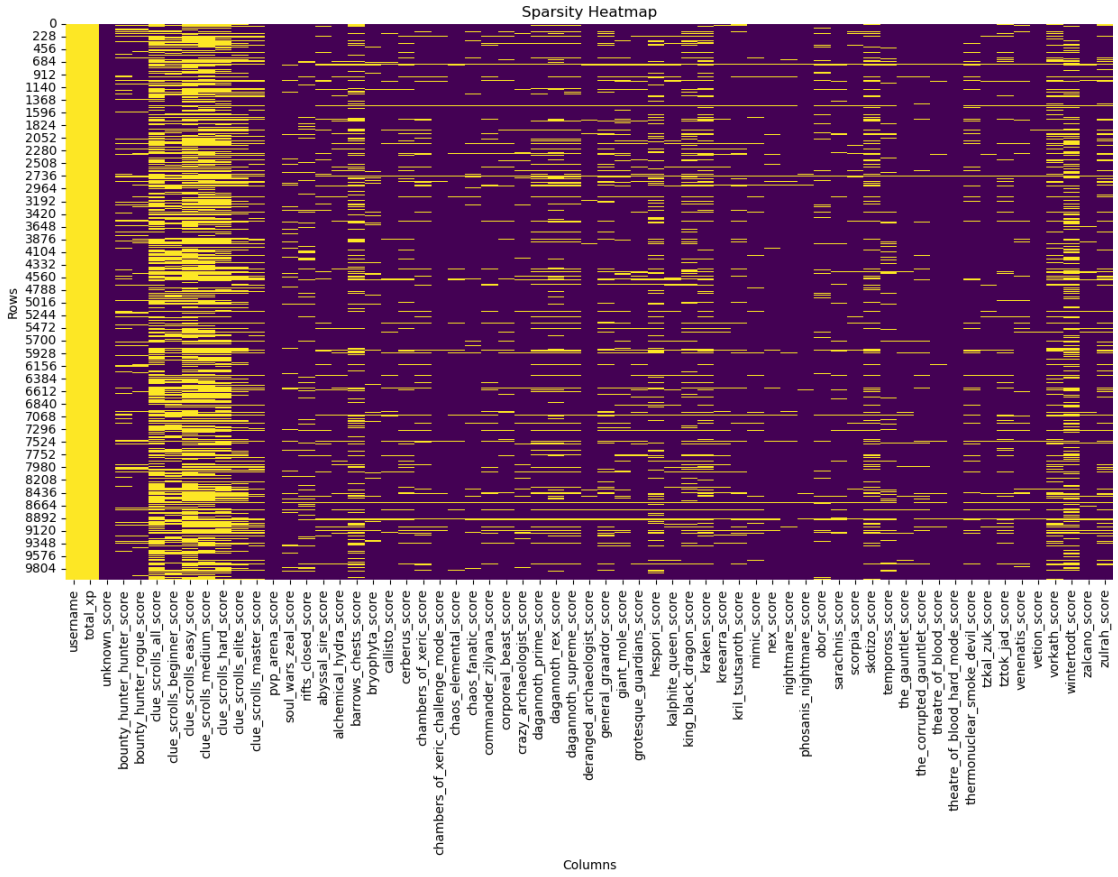
7.1 Purple is NA

As you can see, there is more missing data than avaliable data

[76]:
```
# Convert dataset to a boolean mask (True for non-zero/non-null, False for␣
 ↪sparse)
sparsity_mask = df1.notnull() & (df1 != 0)

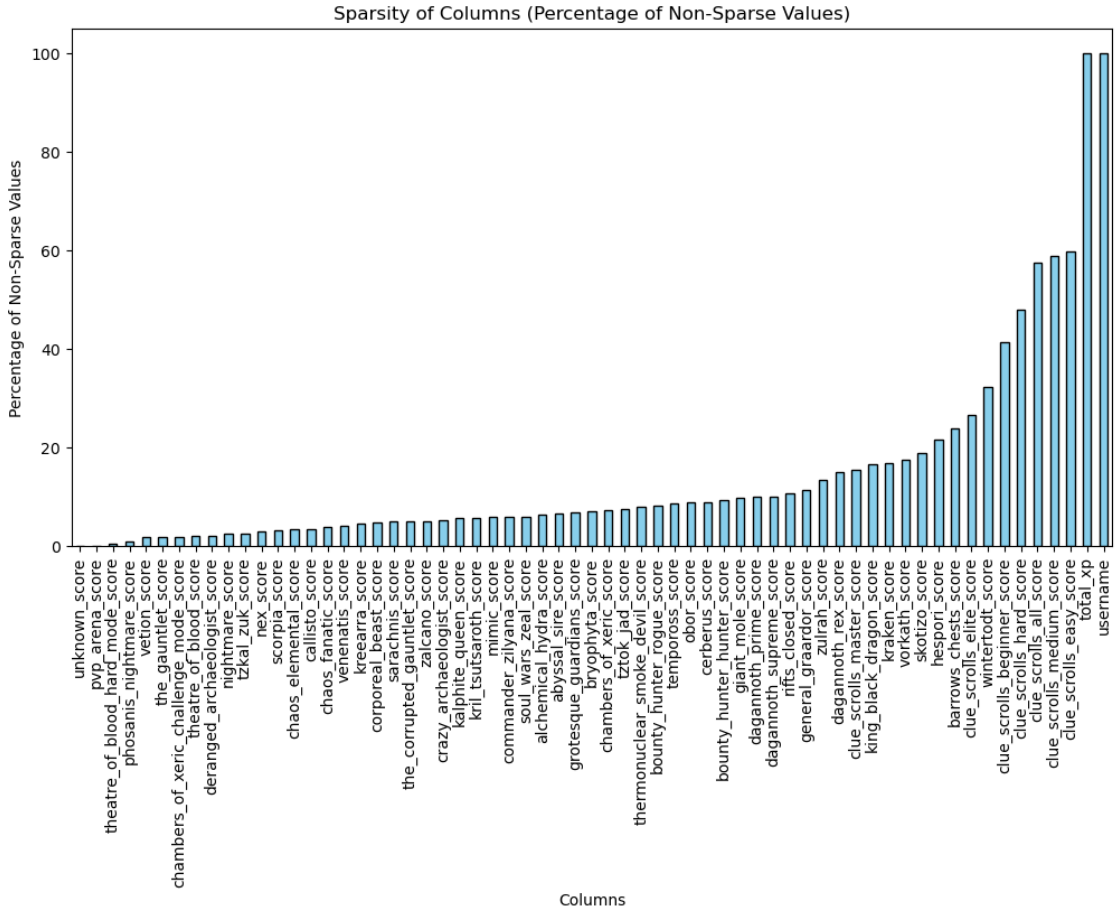
# Plot the heatmap
plt.figure(figsize=(15, 8))
sns.heatmap(sparsity_mask, cbar=False, cmap="viridis")  # 'viridis' for contrast
plt.title("Sparsity Heatmap")
```

```
plt.xlabel("Columns")
plt.ylabel("Rows")
plt.show()
```

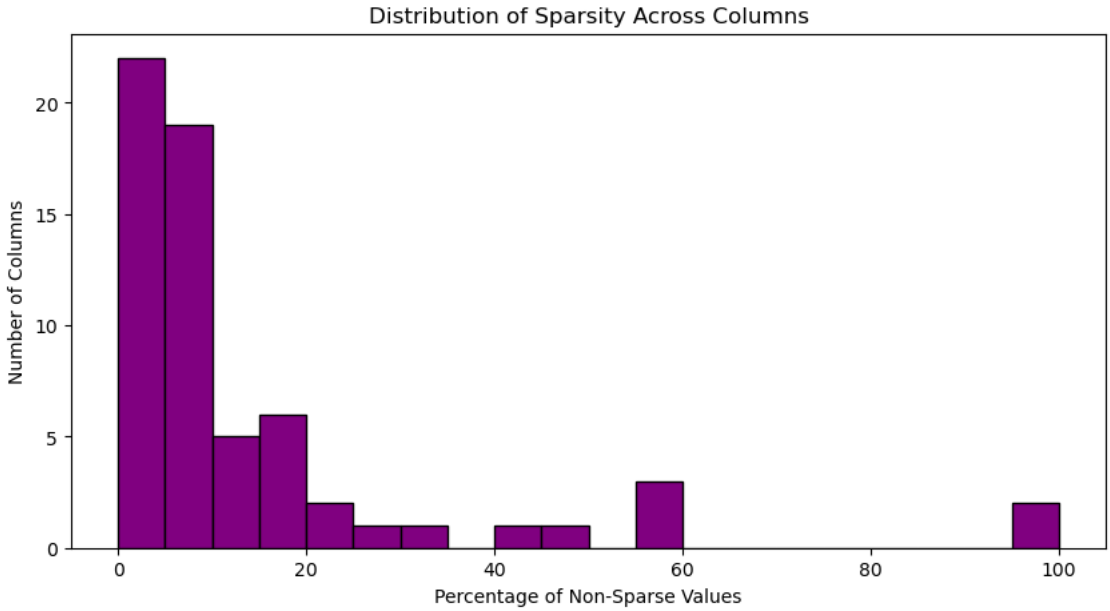


[78]:
```
# Calculate sparsity (percentage of non-null/non-zero values per column)
sparsity = (df1.notnull() & (df1 != 0)).mean() * 100

# Plot
plt.figure(figsize=(12, 6))
sparsity.sort_values().plot(kind='bar', color='skyblue', edgecolor='k')
plt.title("Sparsity of Columns (Percentage of Non-Sparse Values)")
plt.xlabel("Columns")
plt.ylabel("Percentage of Non-Sparse Values")
plt.xticks(rotation=90)
plt.show()
```

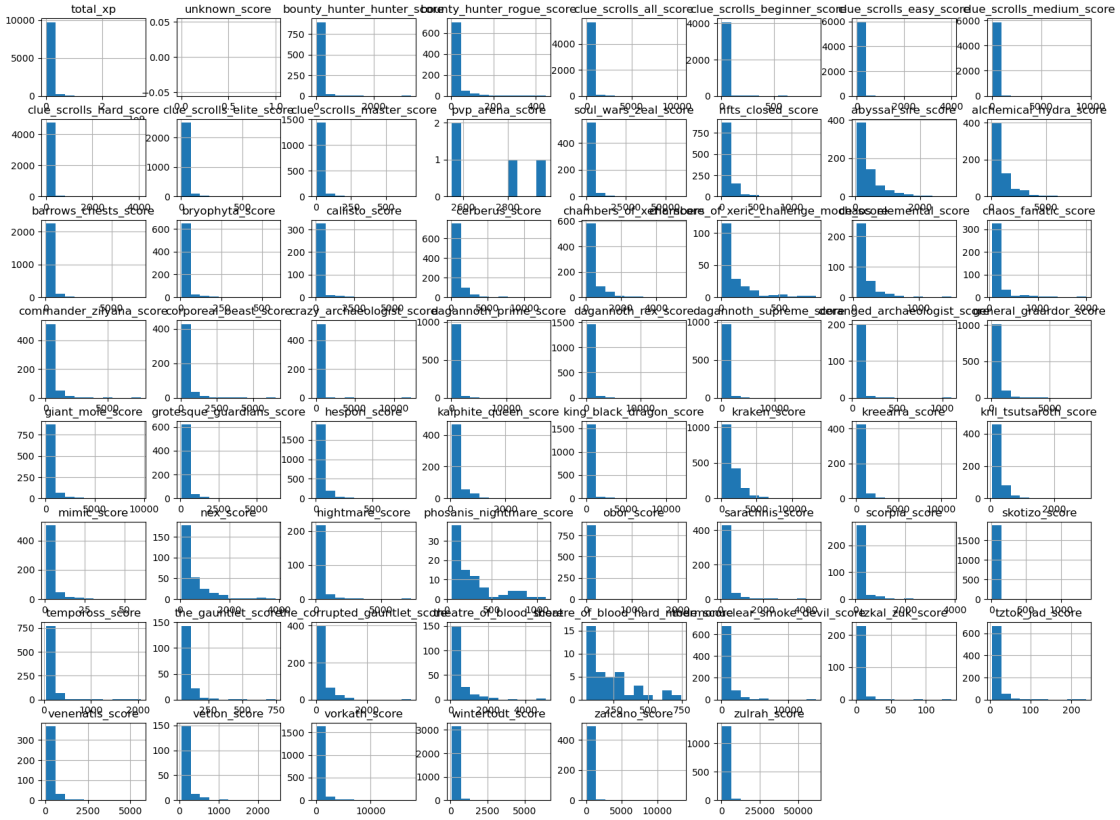


[80]:
```
plt.figure(figsize=(10, 5))
plt.hist(sparsity, bins=20, color='purple', edgecolor='k')
plt.title("Distribution of Sparsity Across Columns")
plt.xlabel("Percentage of Non-Sparse Values")
plt.ylabel("Number of Columns")
plt.show()
```



7.2 Every Single column has an extreme right skew

[88]:
```
df1.hist(figsize = (20,15))
plt.show()
```



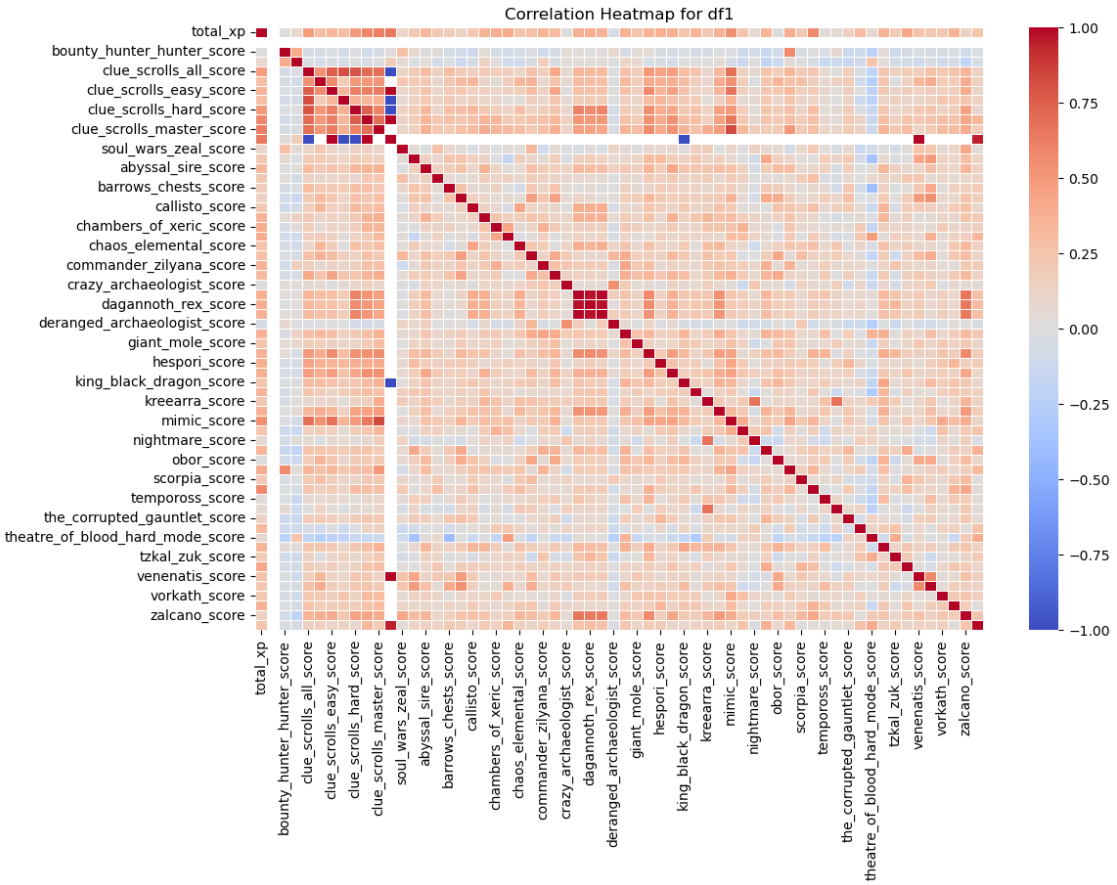
8 Correlation Check

The leftmost column and top most row tell us the most onformation about our expnanatory variable. We can see that there isn't any column with potential data leakage. The highest correlating columns with total_xp are Master Clue Scroll Completion and Mimic Score.

[112]:
```
corr_matrix = df1.corr(numeric_only = True)
plt.figure(figsize=(12, 8))  # Adjust figure size as needed
sns.heatmap(corr_matrix, annot=False, cmap="coolwarm", fmt=".2f", linewidths=0.
↪5)

# Add a title
plt.title("Correlation Heatmap for df1")

# Display the heatmap
plt.show()
```



[116]:
```
df1.nunique()
```

[116]:
```
username                          10000
total_xp                           9998
unknown_score                         0
bounty_hunter_hunter_score          200
bounty_hunter_rogue_score           125
clue_scrolls_all_score              694
clue_scrolls_beginner_score         146
clue_scrolls_easy_score             218
clue_scrolls_medium_score           395
clue_scrolls_hard_score             387
clue_scrolls_elite_score            165
clue_scrolls_master_score           146
pvp_arena_score                       4
soul_wars_zeal_score                495
rifts_closed_score                  257
abyssal_sire_score                  426
alchemical_hydra_score              525
```

```
barrows_chests_score                        651
bryophyta_score                              87
callisto_score                              191
cerberus_score                              623
chambers_of_xeric_score                     443
chambers_of_xeric_challenge_mode_score      118
chaos_elemental_score                       185
chaos_fanatic_score                         173
commander_zilyana_score                     347
corporeal_beast_score                       273
crazy_archaeologist_score                   189
dagannoth_prime_score                       541
dagannoth_rex_score                         641
dagannoth_supreme_score                     547
deranged_archaeologist_score                 43
general_graardor_score                      573
giant_mole_score                            517
grotesque_guardians_score                   338
hespori_score                               201
kalphite_queen_score                        296
king_black_dragon_score                     583
kraken_score                               1258
kreearra_score                              307
kril_tsutsaroth_score                       296
mimic_score                                  25
nex_score                                   239
nightmare_score                             177
phosanis_nightmare_score                     77
obor_score                                   76
sarachnis_score                             233
scorpia_score                               174
skotizo_score                               103
tempoross_score                             249
the_gauntlet_score                           80
the_corrupted_gauntlet_score                284
theatre_of_blood_score                      172
theatre_of_blood_hard_mode_score             40
thermonuclear_smoke_devil_score             588
tzkal_zuk_score                              23
tztok_jad_score                              65
venenatis_score                             247
vetion_score                                101
vorkath_score                               837
wintertodt_score                            672
zalcano_score                               269
zulrah_score                                922
dtype: int64
```

[120]:
```
df1 = df1.drop(columns=['unknown_score'], errors='ignore')
```

[122]:
```
df1.to_csv('OSRSkcDataset.csv', index=False)
```

[ ]:
```

```